

LANSDOWNE HOUSE, ON, Canada

WMO: 718460

Lat: 52.200N

Lon: 87.940W

Elev: 253

StdP: 98.32

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.4	-34.2	-40.9	0.1	-36.6	-38.1	0.1	-33.6	7.8	-13.5	7.2	-14.3	1.4	260	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	28.1	18.6	26.3	17.7	24.5	17.1	20.7	25.4	19.4	23.7	18.3	22.5	4.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.2	23.0	17.8	13.2	21.6	16.7	12.3	20.4	61.0	25.5	56.7	23.8	52.9	22.6	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.9	6.2	DB	-41.2	32.2	3.3	2.4	-43.5	34.0	-45.4	35.4	-47.2	36.7	-49.6	38.4	(4)
(5)			WB	-40.6	22.8	3.0	1.2	-42.7	23.7	-44.5	24.4	-46.2	25.1	-48.3	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-20.7	-18.6	-10.7	-1.6	7.2	14.2	17.1	15.6	10.6	3.0	-6.6	-16.1	(6)	
	DBStd	14.60	7.80	7.34	8.13	6.68	5.62	4.71	3.55	3.69	4.63	4.64	6.96	8.21	(7)	
	HDD10.0	4464	951	799	643	348	124	14	1	2	47	225	499	809	(8)	
	HDD18.3	6922	1209	1033	901	597	347	140	66	102	235	477	749	1067	(9)	
	CDD10.0	649	0	0	1	2	38	139	221	176	66	7	0	0	(10)	
	CDD18.3	66	0	0	0	0	2	15	28	17	3	0	0	0	(11)	
	CDH23.3	647	0	0	0	1	43	179	252	143	29	1	0	0	(12)	
	CDH26.7	137	0	0	0	0	8	42	56	26	4	0	0	0	(13)	
	WSAvg	3.0	2.8	2.7	2.9	3.0	3.2	3.1	2.9	3.0	3.2	3.3	3.0	2.8	(14)	
(15) Precipitation	PrecAvg	719	34	24	32	45	55	85	100	94	90	66	55	35	(15)	
	PrecMax	931	107	58	79	105	93	234	233	216	174	189	107	67	(16)	
	PrecMin	511	10	7	10	7	20	20	39	35	19	16	16	17	(17)	
	PrecStd	113	23	15	16	28	18	45	47	48	40	33	23	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	2.5	12.6	19.3	27.5	30.4	31.1	29.2	26.8	20.0	11.2	2.1	(19)	
		MCWB	0.3	0.9	7.2	10.0	15.2	19.1	20.7	20.4	18.5	14.7	7.8	0.4	(20)	
	2%	DB	-1.9	0.1	7.2	15.1	23.9	27.5	28.0	27.0	23.2	16.1	6.7	0.0	(21)	
		MCWB	-2.4	-1.1	3.1	7.7	13.8	17.4	18.9	19.1	17.2	11.6	4.7	-0.5	(22)	
	5%	DB	-4.6	-2.8	4.3	11.7	20.7	25.4	26.0	24.7	20.6	12.9	4.2	-1.8	(23)	
		MCWB	-5.2	-4.0	1.3	5.3	12.3	16.4	18.1	18.3	15.6	9.9	2.9	-2.4	(24)	
	10%	DB	-8.1	-6.3	1.5	8.6	17.4	23.0	24.2	22.7	18.3	10.1	2.0	-4.2	(25)	
		MCWB	-8.5	-7.2	-0.6	3.5	10.7	15.5	17.3	17.0	14.5	8.1	1.2	-4.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	1.3	7.3	11.1	17.6	21.6	22.5	21.9	19.8	15.6	8.5	0.9	(27)	
		MCDB	1.2	2.2	11.3	18.2	23.6	27.7	28.9	26.7	23.6	19.5	10.3	1.5	(28)	
	2%	WB	-2.4	-1.0	3.6	7.9	14.9	19.4	20.8	20.3	18.1	12.4	4.8	-0.6	(29)	
		MCDB	-1.9	0.1	6.5	14.3	21.5	24.4	25.2	25.0	22.2	14.8	6.2	0.0	(30)	
	5%	WB	-5.1	-3.9	1.6	5.8	13.3	17.8	19.4	19.2	16.4	10.1	2.9	-2.3	(31)	
		MCDB	-4.5	-2.9	4.0	10.9	19.5	22.7	23.6	23.3	19.6	12.4	4.1	-1.8	(32)	
	10%	WB	-8.5	-7.1	-0.4	3.8	11.4	16.4	18.2	17.9	14.7	8.0	1.2	-4.7	(33)	
		MCDB	-8.1	-6.3	1.5	8.1	17.1	21.5	22.6	21.3	17.7	9.9	2.0	-4.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	13.2	14.4	12.8	12.2	11.6	10.9	10.6	9.4	6.9	7.1	9.0	(35)	
		MCDBR	10.6	12.4	14.9	17.2	17.3	15.1	13.9	13.7	13.0	12.2	8.6	8.7	(36)	
		MCWBR	10.2	11.2	11.1	10.6	8.8	7.4	6.9	7.3	7.7	8.4	6.8	8.3	(37)	
	5% WB	MCDBR	10.4	11.7	12.8	15.7	14.8	12.9	11.4	11.4	11.1	11.4	7.9	8.7	(38)	
		MCWBR	10.2	10.7	9.8	10.1	8.3	7.2	6.6	6.8	7.4	8.3	6.5	8.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.237	0.253	0.278	0.332	0.352	0.359	0.388	0.363	0.342	0.297	0.268	0.236	(40)	
	taud		2.301	2.297	2.316	2.346	2.433	2.440	2.391	2.462	2.515	2.579	2.448	2.315	(41)	
	Ebn at Noon		791	882	924	905	900	894	859	865	845	829	751	727	(42)	
	Edh at Noon		60	82	99	111	107	107	111	98	83	64	53	50	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.16	2.40	3.88	5.21	5.23	5.61	5.42	4.41	3.03	1.75	0.90	0.72	(44)	
	RadStd		0.12	0.16	0.30	0.37	0.44	0.63	0.47	0.38	0.33	0.23	0.10	0.11	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page